LabTrack: 15-Slide Presentation Deck Overview & Briefing

Restored to Commit **01db244 / 0eba849** • Pushed to origin/main
Live Preview Servers: **http://127.0.0.1:5174** (Presentation) • **http://localhost:5173** (App)

1. Technical Verification & Architecture Baseline

Repository Status: Clean working tree, synchronised with remote main at commit 0eba849 (restoring 01db244).
Slide Deck Structure: Exactly 15 slides total (Slides 1–13 Core Presentation + Slide 14 Discussion + Slide 15 Academic Citations).
Keynote Animation Physics: Orchestrated CSS build animations using Apple's cubic-bezier(0.16, 1, 0.3, 1) curve. Staggered cascade entrance: T+0.02s gold section badge, T+0.06s slide title, T+0.10s lead thesis paragraph, T+0.14s–T+0.36s sequence/table rows, T+0.32s takeaway banner.
Visual Theme: Luxury dark slate (#0a0a0c) with champagne gold accents (#c9b8a0) and high-contrast typography (58px headings, 24px content text).

2. Slide-by-Slide Technical Specifications & Spoken Pitch

Slide 01: LabTrack (Cover) — Platform Identity

Visual Layout & Content: Title, SIT Pune Dept. of AI & ML, hero microcontroller image, footer status badge: 'AWS + Azure DR • Multi-Stage Docker (:8000) • 61 Verified Tests'.

Core Spoken Pitch & Defense: *"Introduces LabTrack as a hardware lifecycle governance system with tested server workflows, running locally on :5173 and :8000 with a target Azure Container Apps architecture."*

Slide 02: The Problem We Implemented — Problem Formulation

Visual Layout & Content: Three sequence steps: Consumable stock balancing (HC-SR04), Serialized asset custody (SIT barcodes, 1-to-1 logs), and Academic approval authority. Return formula: Q_good + Q_defective = Q_issued.

Core Spoken Pitch & Defense: *"Explains why generic IT asset software fails in academic labs: it lacks capstone project contexts, mentor approvals, and return station condition inspection."*

Slide 03: Literature Review: Service and Deployment — Research Basis

Visual Layout & Content: Comparative table: RFC 5321 (SMTP transport vs outbox registry), FastAPI dependency injection, AWS VPC + private RDS MySQL with Azure DR replica, Docker Compose process packaging (avoids ~$75/mo K8s fee).

Core Spoken Pitch & Defense: *"Establishes how standard RFCs, container isolation, and managed database isolation directly informed LabTrack's architecture choices."*

Slide 04: Literature Review: Correctness and Security — Research Basis

Visual Layout & Content: Comparative table: AWS SQS FIFO (queue ordering vs ACID stock check), PostgreSQL transaction isolation, NIST FIPS 180-4 (SHA-256 integrity hash vs password encryption), Phase 1 security (TLS 1.3, vaults).

Core Spoken Pitch & Defense: *"Clarifies transaction isolation boundaries and proves SHA-256 is an integrity hash rather than password encryption."*

Slide 05: What Runs Today — Current Implementation

Visual Layout & Content: Empirical status table: React+Vite on :5173, FastAPI on :8000 in Docker, SQLite business state (73 users, 15 projects, 12 requests), reachable MongoDB Atlas projections, CCE presentation on :5174/:5175.

Core Spoken Pitch & Defense: *"Demonstrates absolute engineering honesty: discloses what runs live on localhost versus what is designed for target cloud deployment."*

Slide 06: The Implementation Boundary — System Boundary

Visual Layout & Content: Four architecture cards: Visual Workstation (React UI) -> Authoritative Service (FastAPI) -> Current Gap (two state stores) -> Demonstration Method (UI for operator flow, API for mathematical proof).

Core Spoken Pitch & Defense: *"Preempts examiner questions by clearly identifying that UI currently uses simulated calls while the backend is an independently tested, authoritative SQL service."*

Slide 07: Backend Workflow Implemented — Server Logic

Visual Layout & Content: Six-step state machine: 01 Authenticate (JWT Bearer) -> 02 Create project -> 03 Plan & submit (minimum-lab allocation) -> 04 Approve by role -> 05 Claim & issue -> 06 Inspect return.

Core Spoken Pitch & Defense: *"Proves the backend enforces rigorous state transitions, role authorizations, and stock reconciliation math at every step of the hardware lifecycle."*

Slide 08: Inventory Rules in Code — Server Logic

Visual Layout & Content: Split inventory cards with hardware images: Consumables (HC-SR04 ultrasonic sensor, Q_good + Q_defective = Q_issued) vs Serialized Assets (robotic arm, barcode SIT-ROBOT-0042, AVAILABLE -> RESERVED -> ISSUED -> RETURNED).

Core Spoken Pitch & Defense: *"Highlights distinct data models: consumables track aggregate quantities per lab, while assets operate on a 1-to-1 physical custody state machine."*

Slide 09: Database and Event Handling — Cloud Implementation

Visual Layout & Content: Comparative table: Relational data (SQLAlchemy + RDS MySQL Multi-AZ & Azure DR) -> Notifications (MongoDB Atlas) -> Email (AWS SQS FIFO + outbox worker) -> Repeat commands (Idempotency keys).

Core Spoken Pitch & Defense: *"Details polyglot persistence and the Transactional Outbox pattern: SQL commits business state first; asynchronous workers dispatch emails without blocking checkout."*

Slide 10: Verified Backend Evidence — Validation

Visual Layout & Content: Metric card: '61 passed', 2 skipped, 0 failures, 397.77s duration. Docker container: non-root uid 10001, healthcheck PASS. Test coverage list: JWT auth, planning, 403 Forbidden checks, atomic issue/return, idempotency.

Core Spoken Pitch & Defense: *"Primary empirical evidence slide proving robust server logic, automated regression prevention, and container security compliance."*

Slide 11: Live Demonstration — Running Example

Visual Layout & Content: Four-step execution flow with direct clickable buttons: Open website (:5173) and Open API documentation (:8000/docs). Demo credentials: DEMO-STU-001 / DemoOnly2026!.

Core Spoken Pitch & Defense: *"Transition slide for the 4-minute live demonstration: showing the visual operator interface on 5173 and server-side Swagger execution on 8000."*

Slide 12: One Failure Case — Running Example

Visual Layout & Content: Three graceful rejection mechanisms: Shortage refusal (422 Unprocessable Entity), Unauthorized approver (403 Forbidden), and Concurrency race refusal (409 Conflict rollback).

Core Spoken Pitch & Defense: *"Proves system resilience and negative testing: the database and server defend business rules when malformed or illegal requests are sent."*

Slide 13: Known Integration Work — Engineering Status

Visual Layout & Content: Prioritized roadmap table: P1 backend login, P1 replace browser writes with FastAPI calls, P1 outbox delivery enforcement, P1 server return reconciliation, P2 Docker Compose cloud deploy.

Core Spoken Pitch & Defense: *"Demonstrates engineering maturity by presenting a realistic, professional engineering backlog moving LabTrack toward production."*

Slide 14: Discussion — Viva Defense

Visual Layout & Content: Closing statement: 'The implementation exists. The interface integration is next.' Four discussion prompts: Why SQL and MongoDB? Why an outbox? What protects stock? What integrates next?

Core Spoken Pitch & Defense: *"Guides the evaluation committee into asking questions the student has thoroughly prepared for."*

Slide 15: References and Evidence — Appendix

Visual Layout & Content: Nine formal academic citations: RFC 5321 (SMTP), FastAPI Background Tasks, OWASP FastAPI Security, AWS SQS FIFO, PostgreSQL Isolation, AWS EC2/RDS/Docker, NIST FIPS 180-4, SDD & Test Report, and LITERATURE.md.

Core Spoken Pitch & Defense: *"Anchors every architectural claim in industry standards and official academic specifications."*